main.c

```c
#include <stdio.h>

#define MAX 5

int pq[MAX];
int size = 0;

// Insert element in sorted order
void enqueue(int value)
{
    int i;

    if(size == MAX)
    {
        printf("Priority Queue Overflow\n");
        return;
    }

    // Insert based on priority (smaller number = higher priority)
    for(i = size - 1; i >= 0 && pq[i] > value; i--)
    {
        pq[i + 1] = pq[i];
    }

    pq[i + 1] = value;
    size++;

    printf("%d inserted\n", value);
}
```

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```c
// Delete highest priority element
void dequeue()
{
    int i;

    if(size == 0)
    {
        printf("Priority Queue Underflow\n");
        return;
    }

    printf("%d deleted\n", pq[0]);

    for(i = 0; i < size - 1; i++)
    {
        pq[i] = pq[i + 1];
    }

    size--;
}

// Display elements
void display()
{
    int i;

    if(size == 0)
    {
        printf("Priority Queue is Empty\n");
    }
```

main.c

```c
        printf("Priority Queue Elements:\n");

        for(i = 0; i < size; i++)
        {
            printf("%d ", pq[i]);
        }
    }

    printf("\n");
}

int main()
{
    enqueue(30);
    enqueue(10);
    enqueue(50);
    enqueue(20);

    display();

    dequeue();

    display();

    return 0;
}
```

"E:\C-Programming\DS progr

```
30 inserted
10 inserted
50 inserted
20 inserted
Priority Queue Elements:
10 20 30 50
10 deleted
Priority Queue Elements:
20 30 50

Process returned 0 (0x0)   execution time : 0.114 s
Press any key to continue.
```

Code::Blocks X Search results X Cccc X Build log X Build messages X CppCheck/Vera++ X CppCheck/Vera++ messages X Cscope X Debugger X DoxyBlocks X Fortran info X Clo

```
-------------- Run: Debug in DS program 8 (compiler: GNU GCC Compiler
Checking for existence: E:\C-Programming\
Set variable: PATH=.;C:\
```